Supplementary Information
MethylLlama: self-supervised foundation modeling of human DNA methylation profiles

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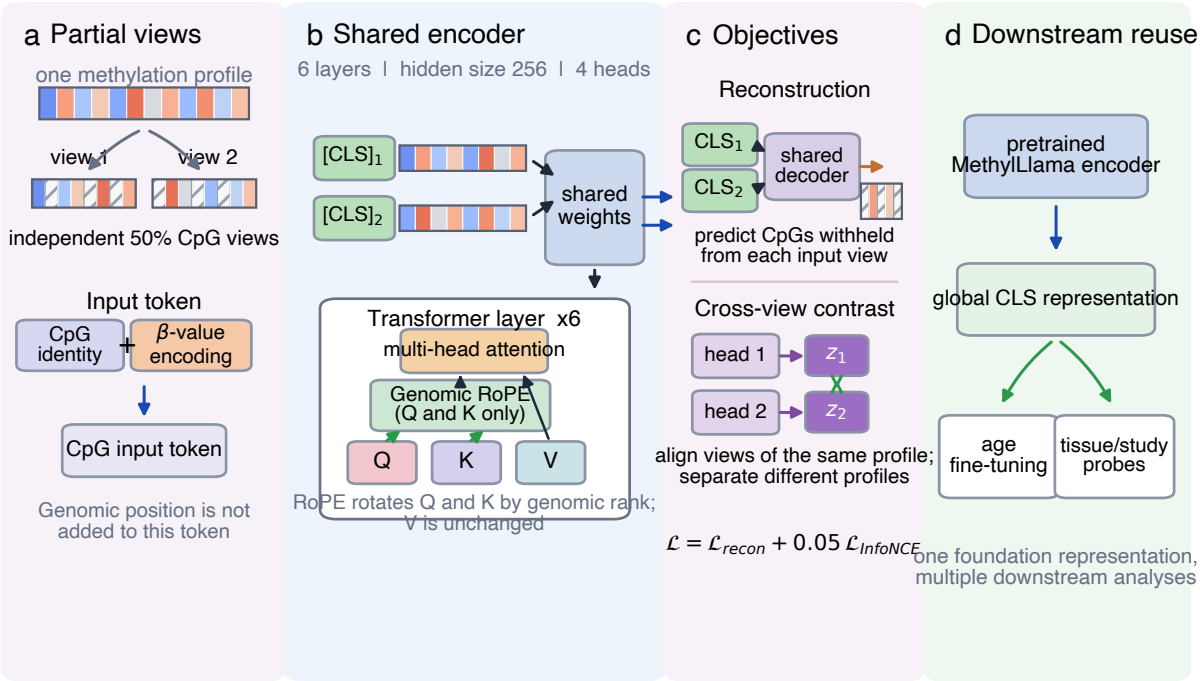


Figure S1: MethylLlama architecture and the WCED pretraining objective. **a**, Partial-view inputs and token construction. Two independently sampled 50% CpG views are drawn from each profile. Each input token combines a learned CpG-identity embedding with a continuous β-value encoding; genomic position is not added to the token. **b**, Shared encoder. Six Transformer layers of hidden size 256 with four attention heads process each view alongside a prepended CLS token. Genomic rank enters exclusively as rotary position applied to the query and key projections inside self-attention; the value projection is unchanged. **c**, Dual self-supervised objectives. A shared decoder predicts CpGs withheld from each view using its CLS representation, while symmetric InfoNCE aligns the two views of one profile against other profiles in the batch. **d**, Downstream reuse of the pretrained encoder for chronological-age fine-tuning and frozen tissue and study probes. Schematic only; no measured data are plotted.

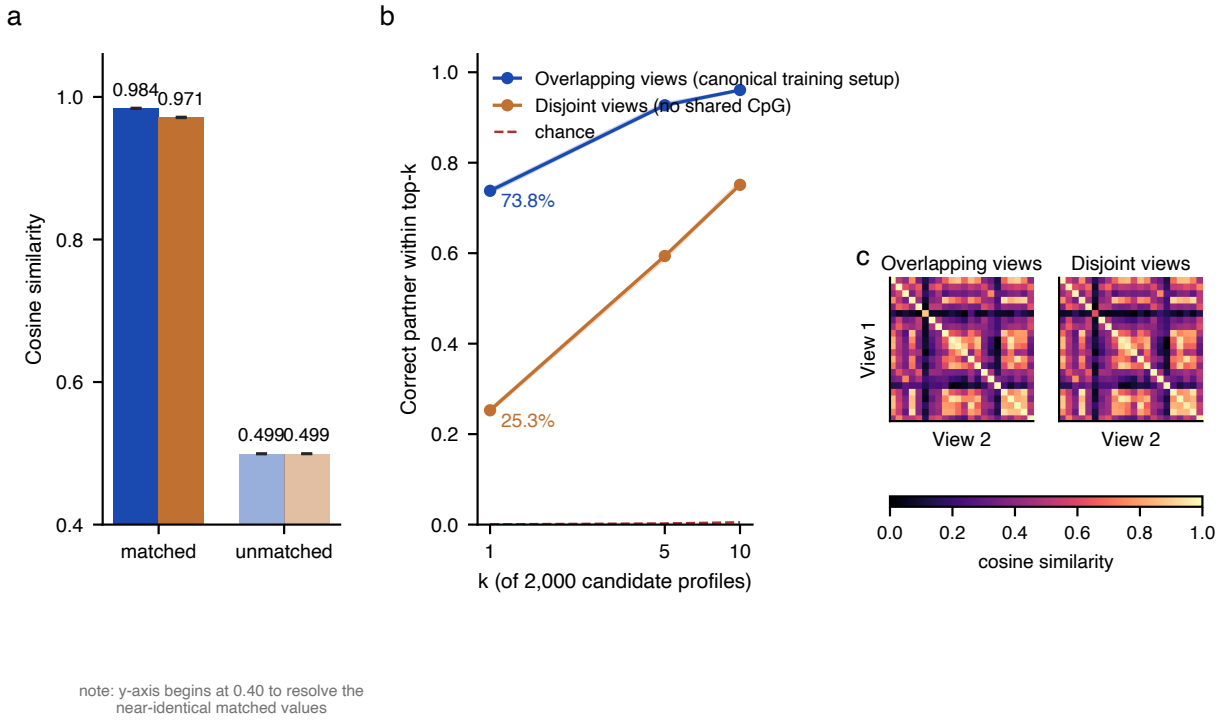


Figure S2: Cross-view consistency depends partly on overlap between the two views. Two view constructions were compared on the same 2,000 profiles using the same pretrained encoder: the canonical training construction, in which the two 50% CpG subsets are drawn independently and therefore share approximately half of their sites, and a disjoint construction, in which the subsets partition the valid CpGs and share none. **a**, Mean cosine similarity for matched and unmatched pairs. Bars show the mean and error bars the standard deviation across five independent random view draws. The vertical axis begins at 0.40 so that the near-identical matched values remain distinguishable. **b**, Retrieval accuracy as a function of the number of candidates considered. Lines show the mean and shading the minimum–maximum range across the same five draws; the dashed line is chance. **c**, Similarity matrices for one draw under each construction (22 profiles drawn at random under a fixed seed, shared colour scale). Matched similarity is nearly unchanged between constructions (0.984 versus 0.971) whereas top-1 retrieval falls from 73.8% to 25.3%, converging again as more candidates are considered. Because the encoder was trained using independently sampled, potentially overlapping views, the disjoint condition is out of distribution for it, and these values are a conservative bound on cross-subset consistency rather than a matched comparison.

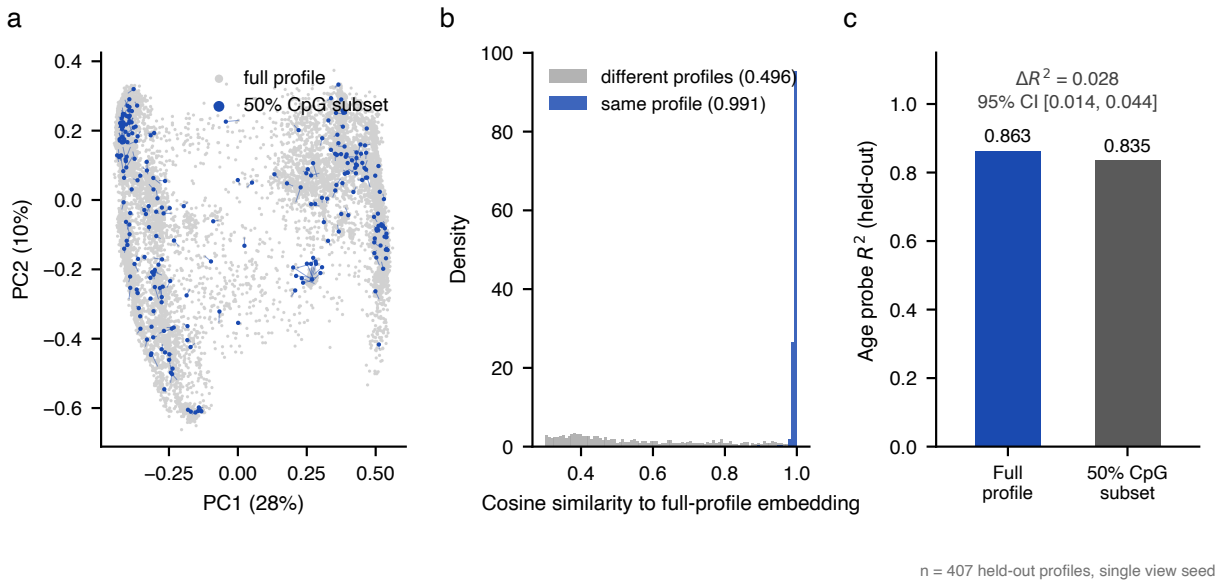


Figure S3: Robustness of pretrained representations to random CpG subsampling. a, Complete profiles (grey) and 50% CpG subsets of the same profiles (blue) projected into a shared principal-component space fitted on the complete profiles, with a segment joining each of 250 randomly chosen profiles to its own subset. **b**, Per-profile cosine similarity between the complete-profile embedding and the embedding of a 50% subset of the same profile, against a shuffled-pairing control. **c**, A ridge age probe fitted on complete-profile embeddings of training profiles and applied, without adaptation, to complete and to 50%-subset embeddings of 407 held-out profiles; the annotated interval is a paired bootstrap 95% confidence interval on the difference over 5,000 resamples of those profiles. Subsets are random 50% draws from a single view seed, not the probe set of a specific array generation, so this experiment does not establish independence from a real measurement platform. Performance is not identical: the probe falls from $R^2 = 0.863$ to $R^2 = 0.835$, a difference whose confidence interval excludes zero. The result therefore indicates substantial robustness to which CpGs are observed, not equivalence.

Table S1: Per-partition age-prediction metrics on the common 2,149-sample test set. Values were recomputed from saved sample-level predictions rather than read from training logs. Every model was evaluated on the same test samples, so the dispersion across partitions quantifies sensitivity to the supervised training partition and is not a population confidence interval. The fold ensemble averages the five per-sample predictions.

Training partition	MedAE (years)	MAE (years)	R^2
Partition 0	3.126	4.439	0.9321
Partition 1	3.213	4.458	0.9337
Partition 2	3.209	4.570	0.9284
Partition 3	3.152	4.541	0.9298
Partition 4	3.144	4.455	0.9338
Mean ± SD	3.169 ± 0.040	4.493 ± 0.059	0.9316 ± 0.0024
Fold ensemble	3.112	4.333	0.9365

Table S2: Geometry of the frozen pretrained representation. Computed over all 10,988 profiles of the AltuMAge collection using the pretrained encoder without fine-tuning. Effective rank is the exponential of the entropy of the normalized singular-value spectrum; anisotropy is the mean pairwise cosine similarity, for which lower values indicate a less directionally collapsed space. The CLS representation occupies a substantially higher-dimensional and less anisotropic subspace than uniform pooling over contextual CpG tokens of the same model.

	CLS representation	Mean-pooled CpG tokens
Effective rank	18.12	3.40
Mean cosine anisotropy	0.500	0.867
Mean L_2 norm	10.01	5.36

Supplementary Note 1: genomic-rank rotary position encoding

To probe the representational effect of supplying genomic rather than sequential positions to the rotary position encoding, we trained a matched pilot pair on the same fixed 5,000-profile subset. The two runs shared architecture, objective, and hyperparameters; the intended difference was whether rotary position encoding received genomic-rank or sequential position indices.

For the genomic-rank model, the mean cosine similarity of contextualized CpG representations was 0.9831 for genomically adjacent CpGs and 0.8857 for random pairs, a locality gap of 0.0974. Without genomic positions the corresponding values were 0.9931 and 0.9621, a gap of 0.0310. The separation of adjacent from random CpGs was therefore approximately 3.14-fold larger with genomic positions. In the production model's static token-embedding table, adjacent and random CpGs showed no positive locality gap, which is consistent with locality arising through contextual computation rather than through clustering of identity embeddings.

This is a pilot and is reported as such. It used a single seed, and the selected genomic-rank checkpoint also had better reconstruction diagnostics than the non-genomic checkpoint (normalized reconstruction loss 0.0906 versus 0.1585; Pearson correlation 0.9357 versus 0.9017), so the two runs are not matched on training quality and the locality difference cannot be cleanly attributed to the position encoding alone. We make no claim that genomic-rank positions improve downstream predictive performance; establishing that would require replicated pretraining runs and a matched downstream ablation.